Principle Component Analysis

Principal Component Analysis (PCA) is a linear dimensionality reduction technique that can be utilized for extracting information from a high-dimensional space by projecting it into a lower-dimensional sub-space.

1. It tries to preserve the essential parts that have more variation of the data and remove the non-essential parts with fewer variation.
2. One important thing to note about PCA is that it is an Unsupervised dimensionality reduction technique, you can cluster the similar data points based on the feature correlation between them without any supervision (or labels)
3. PCA is a statistical procedure that uses an orthogonal transformation to convert a set of observations of possibly correlated variables (entities each of which takes on various numerical values) into a set of values of linearly uncorrelated variables called principal components.

Note: Features, Dimensions, and Variables are all referring to the same thing.

4. Used in Data Visualization for dimensionality reduction and in ML for speeding up the algorithm.
5. When the data is projected into a lower dimension (assume three dimensions) from a higher space, the three dimensions are nothing but the three Principal Components that captures (or holds) most of the variance (information) of your data.

Principal components have both direction and magnitude.

- The direction represents across which principal axes the data is mostly spread out or has most variance.
- The magnitude signifies the amount of variance that Principal Component captures of the data when projected onto that axis.

Here we are going to use Breast Cancer dataset. It contains two classes of data whether the patient has a Breast cancer or not. The two classes are **Malignant** and **Benign**

```
import numpy as np
import pandas as pd
import matplotlib as plt
from sklearn.datasets import load_breast_cancer
```

load_breast_cancer will give you both labels and the data. To fetch the data, you will call .data and for fetching the labels .target.

```
cancer = load_breast_cancer()
cancer_data = cancer.data
print(cancer_data)
print(cancer_data.shape)
```

```
[[1.799e+01 1.038e+01 1.228e+02 ... 2.654e-01 4.601e-01 1.189e-01]
 [2.057e+01 1.777e+01 1.329e+02 ... 1.860e-01 2.750e-01 8.902e-02]
 [1.969e+01 2.125e+01 1.300e+02 ... 2.430e-01 3.613e-01 8.758e-02]
 ...
 [1.660e+01 2.808e+01 1.083e+02 ... 1.418e-01 2.218e-01 7.820e-02]
 [2.060e+01 2.933e+01 1.401e+02 ... 2.650e-01 4.087e-01 1.240e-01]
 [7.760e+00 2.454e+01 4.792e+01 ... 0.000e+00 2.871e-01 7.039e-02]]
(569, 30)
```

```
cancer_lables=cancer.target
print(cancer_lables)
print(cancer_lables.shape)
```

```
[0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 1 0 0 0 0 0 0 0 0 1 0 1 1 1 1 1 0 0 1 0 0 1 1 1 1 0 1 0 0 1 1 1 1 0 1 0 0
 1 0 1 0 0 1 1 1 0 0 1 0 0 0 1 1 1 0 1 1 0 0 1 1 1 0 0 1 1 1 1 0 1 1 0 1 1
 1 1 1 1 1 1 0 0 0 1 0 0 1 1 1 0 0 1 0 1 0 0 1 0 0 1 1 0 1 1 0 1 1 1 1 0 1
 1 1 1 1 1 1 1 1 0 1 1 1 1 0 0 1 0 1 1 0 0 1 1 0 0 1 1 1 1 0 1 1 0 0 0 1 0
 1 0 1 1 1 0 1 1 0 0 1 0 0 0 0 1 0 0 0 1 0 1 0 1 1 0 1 0 0 0 0 1 1 0 0 1 1
 1 0 1 1 1 1 1 0 0 1 1 0 1 1 0 0 1 0 1 1 1 1 0 1 1 1 1 1 0 1 0 0 0 0 0 0 0
 0 0 0 0 0 0 0 1 1 1 1 1 1 0 1 0 1 1 0 1 1 0 1 0 0 1 1 1 1 1 1 1 1 1 1 1 1
 1 0 1 1 0 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 1 1 1 0 1 0 1 1 1 0 0 0 1 1
 1 1 0 1 0 1 0 1 1 1 0 1 1 1 1 1 1 1 0 0 0 1 1 1 1 1 1 1 1 1 1 0 0 1 0 0
 0 1 0 0 1 1 1 1 1 0 1 1 1 1 1 0 1 1 1 0 1 1 0 0 1 1 1 1 1 1 0 1 1 1 1 1 1
 1 0 1 1 1 1 1 0 1 1 0 1 1 1 1 1 1 1 1 1 1 1 0 1 0 0 1 0 1 1 1 1 1 0 1 1
 0 1 0 1 1 0 1 0 1 1 1 1 1 1 1 1 0 0 1 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1 0 1
 1 1 1 1 1 1 0 1 0 1 1 0 1 1 1 1 1 0 0 1 0 1 0 1 1 1 1 1 0 1 1 0 1 0 1 0 0
 1 1 1 0 1 1 1 1 1 1 1 1 1 1 1 0 1 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
 1 1 1 1 1 1 1 0 0 0 0 0 0 1]
(569,)
```

Reshape the dataset by adding lable to it.

Concatenate the dataset with the lable

```
labels = np.reshape(cancer_lables,(569,1))
final_cancer_data = np.concatenate([cancer_data,labels],axis=1)
print(final_cancer_data.shape)
```

```
(569, 31)
```

Create a dataframe

```
cancer_dataset = pd.DataFrame(final_cancer_data)
print(cancer_dataset.head())
```

```
       0      1       2       3        4        5       6        7       8   \
0  17.99  10.38  122.80  1001.0  0.11840  0.27760  0.3001  0.14710  0.2419
1  20.57  17.77  132.90  1326.0  0.08474  0.07864  0.0869  0.07017  0.1812
2  19.69  21.25  130.00  1203.0  0.10960  0.15990  0.1974  0.12790  0.2069
3  11.42  20.38   77.58   386.1  0.14250  0.28390  0.2414  0.10520  0.2597
4  20.29  14.34  135.10  1297.0  0.10030  0.13280  0.1980  0.10430  0.1809

         9   ...     21      22      23      24      25      26      27  \
0  0.07871  ...  17.33  184.60  2019.0  0.1622  0.6656  0.7119  0.2654
1  0.05667  ...  23.41  158.80  1956.0  0.1238  0.1866  0.2416  0.1860
2  0.05999  ...  25.53  152.50  1709.0  0.1444  0.4245  0.4504  0.2430
3  0.09744  ...  26.50   98.87   567.7  0.2098  0.8663  0.6869  0.2575
4  0.05883  ...  16.67  152.20  1575.0  0.1374  0.2050  0.4000  0.1625

       28       29   30
0  0.4601  0.11890  0.0
1  0.2750  0.08902  0.0
2  0.3613  0.08758  0.0
3  0.6638  0.17300  0.0
4  0.2364  0.07678  0.0

[5 rows x 31 columns]
```

print the features that are there in the breast cancer dataset!

```
features = cancer.feature_names
print(features)
```

```
['mean radius' 'mean texture' 'mean perimeter' 'mean area'
 'mean smoothness' 'mean compactness' 'mean concavity'
 'mean concave points' 'mean symmetry' 'mean fractal dimension'
 'radius error' 'texture error' 'perimeter error' 'area error'
 'smoothness error' 'compactness error' 'concavity error'
 'concave points error' 'symmetry error' 'fractal dimension error'
 'worst radius' 'worst texture' 'worst perimeter' 'worst area'
 'worst smoothness' 'worst compactness' 'worst concavity'
 'worst concave points' 'worst symmetry' 'worst fractal dimension']
```

Here the label field is missing so add to it

```
features_labels = np.append(features,'label')
```

Embedding the column names to the dataframe

```
cancer_dataset.columns = features_labels
cancer_dataset.head()
```

	mean radius	mean texture	mean perimeter	mean area	mean smoothness	mean compactness	mean concavity	mean concave points	mean symmetry	mean fractal dimension	...	worst texture	worst perimeter	wor ar
0	17.99	10.38	122.80	1001.0	0.11840	0.27760	0.3001	0.14710	0.2419	0.07871	...	17.33	184.60	201
1	20.57	17.77	132.90	1326.0	0.08474	0.07864	0.0869	0.07017	0.1812	0.05667	...	23.41	158.80	195
2	19.69	21.25	130.00	1203.0	0.10960	0.15990	0.1974	0.12790	0.2069	0.05999	...	25.53	152.50	170
3	11.42	20.38	77.58	386.1	0.14250	0.28390	0.2414	0.10520	0.2597	0.09744	...	26.50	98.87	56
4	20.29	14.34	135.10	1297.0	0.10030	0.13280	0.1980	0.10430	0.1809	0.05883	...	16.67	152.20	157

5 rows × 31 columns

Since the original labels are in 0,1 format, you will change the labels to benign and malignant using **.replace** function. You will use **inplace=True** which will modify the dataframe breast_dataset.

```
cancer_dataset['label'].replace(0, 'Benign',inplace=True)
cancer_dataset['label'].replace(1, 'Malignant',inplace=True)
cancer_dataset.tail()
```

	mean radius	mean texture	mean perimeter	mean area	mean smoothness	mean compactness	mean concavity	mean concave points	mean symmetry	mean fractal dimension	...	worst texture	worst perimeter	w
564	21.56	22.39	142.00	1479.0	0.11100	0.11590	0.24390	0.13890	0.1726	0.05623	...	26.40	166.10	2
565	20.13	28.25	131.20	1261.0	0.09780	0.10340	0.14400	0.09791	0.1752	0.05533	...	38.25	155.00	1
566	16.60	28.08	108.30	858.1	0.08455	0.10230	0.09251	0.05302	0.1590	0.05648	...	34.12	126.70	1
567	20.60	29.33	140.10	1265.0	0.11780	0.27700	0.35140	0.15200	0.2397	0.07016	...	39.42	184.60	1
568	7.76	24.54	47.92	181.0	0.05263	0.04362	0.00000	0.00000	0.1587	0.05884	...	30.37	59.16	

5 rows × 31 columns

PCA

You start by Standardizing the data since PCA's output is influenced based on the scale of the features of the data.

```
from sklearn.preprocessing import StandardScaler
x = cancer_dataset.loc[:, features].values
x = StandardScaler().fit_transform(x) # normalizing the features
print(x.shape)
```

```
(569, 30)
```

Check the mean and std deviation is 0 and 1

```
np.mean(x),np.std(x)
```

```
(-6.118909323768877e-16, 1.0)
```

Let's convert the normalized features into a tabular format with the help of DataFrame.

```
feat_cols = ['feature'+str(i) for i in range(x.shape[1])]
```

```
normalised_cancer = pd.DataFrame(x,columns=feat_cols)
print(normalised_cancer)
```

```
     feature0  feature1  feature2  feature3  feature4  feature5  feature6  \
0    1.097064 -2.073335  1.269934  0.984375  1.568466  3.283515  2.652874
1    1.829821 -0.353632  1.685955  1.908708 -0.826962 -0.487072 -0.023846
2    1.579888  0.456187  1.566503  1.558884  0.942210  1.052926  1.363478
3   -0.768909  0.253732 -0.592687 -0.764464  3.283553  3.402909  1.915897
4    1.750297 -1.151816  1.776573  1.826229  0.280372  0.539340  1.371011
..        ...       ...       ...       ...       ...       ...       ...
564  2.110995  0.721473  2.060786  2.343856  1.041842  0.219060  1.947285
565  1.704854  2.085134  1.615931  1.723842  0.102458 -0.017833  0.693043
566  0.702284  2.045574  0.672676  0.577953 -0.840484 -0.038680  0.046588
567  1.838341  2.336457  1.982524  1.735218  1.525767  3.272144  3.296944
568 -1.808401  1.221792 -1.814389 -1.347789 -3.112085 -1.150752 -1.114873

     feature7  feature8  feature9  ...  feature20  feature21  feature22  \
0    2.532475  2.217515  2.255747  ...   1.886690  -1.359293   2.303601
1    0.548144  0.001392 -0.868652  ...   1.805927  -0.369203   1.535126
2    2.037231  0.939685 -0.398008  ...   1.511870  -0.023974   1.347475
3    1.451707  2.867383  4.910919  ...  -0.281464   0.133984  -0.249939
4    1.428493 -0.009560 -0.562450  ...   1.298575  -1.466770   1.338539
..        ...       ...       ...  ...        ...        ...        ...
564  2.320965 -0.312589 -0.931027  ...   1.901185   0.117700   1.752563
565  1.263669 -0.217664 -1.058611  ...   1.536720   2.047399   1.421940
566  0.105777 -0.809117 -0.895587  ...   0.561361   1.374854   0.579001
567  2.658866  2.137194  1.043695  ...   1.961239   2.237926   2.303601
568 -1.261820 -0.820070 -0.561032  ...  -1.410893   0.764190  -1.432735

     feature23  feature24  feature25  feature26  feature27  feature28  \
0     2.001237   1.307686   2.616665   2.109526   2.296076   2.750622
1     1.890489  -0.375612  -0.430444  -0.146749   1.087084  -0.243890
2     1.456285   0.527407   1.082932   0.854974   1.955000   1.152255
3    -0.550021   3.394275   3.893397   1.989588   2.175786   6.046041
4     1.220724   0.220556  -0.313395   0.613179   0.729259  -0.868353
..         ...        ...        ...        ...        ...        ...
564   2.015301   0.378365  -0.273318   0.664512   1.629151  -1.360158
```

```
565   1.494959  -0.691230  -0.394820   0.236573   0.733827  -0.531855
566   0.427906  -0.809587   0.350735   0.326767   0.414069  -1.104549
567   1.653171   1.430427   3.904848   3.197605   2.289985   1.919083
568  -1.075813  -1.859019  -1.207552  -1.305831  -1.745063  -0.048138

     feature29
0     1.937015
1     0.281190
2     0.201391
3     4.935010
4    -0.397100
..         ...
564  -0.709091
565  -0.973978
566  -0.318409
567   2.219635
568  -0.751207

[569 rows x 30 columns]
```

```
normalised_cancer.tail()
```

	feature0	feature1	feature2	feature3	feature4	feature5	feature6	feature7
564	2.110995	0.721473	2.060786	2.343856	1.041842	0.219060	1.947285	2.320965
565	1.704854	2.085134	1.615931	1.723842	0.102458	-0.017833	0.693043	1.263669
566	0.702284	2.045574	0.672676	0.577953	-0.840484	-0.038680	0.046588	0.105777
567	1.838341	2.336457	1.982524	1.735218	1.525767	3.272144	3.296944	2.658866
568	-1.808401	1.221792	-1.814389	-1.347789	-3.112085	-1.150752	-1.114873	-1.261820

5 rows × 30 columns

Projecting the thirty-dimensional Breast Cancer data to two-dimensional principal components.

```
from sklearn.decomposition import PCA
pca_cancer = PCA(n_components=2)
principalComponents_cancer = pca_cancer.fit_transform(x)
```

```
principal_cancer_Df = pd.DataFrame(data = principalComponents_cancer
             , columns = ['principal component 1', 'principal component 2'])
principal_cancer_Df.tail()
```

	principal component 1	principal component 2
564	6.439315	-3.576817
565	3.793382	-3.584048
566	1.256179	-1.902297
567	10.374794	1.672010
568	-5.475243	-0.670637

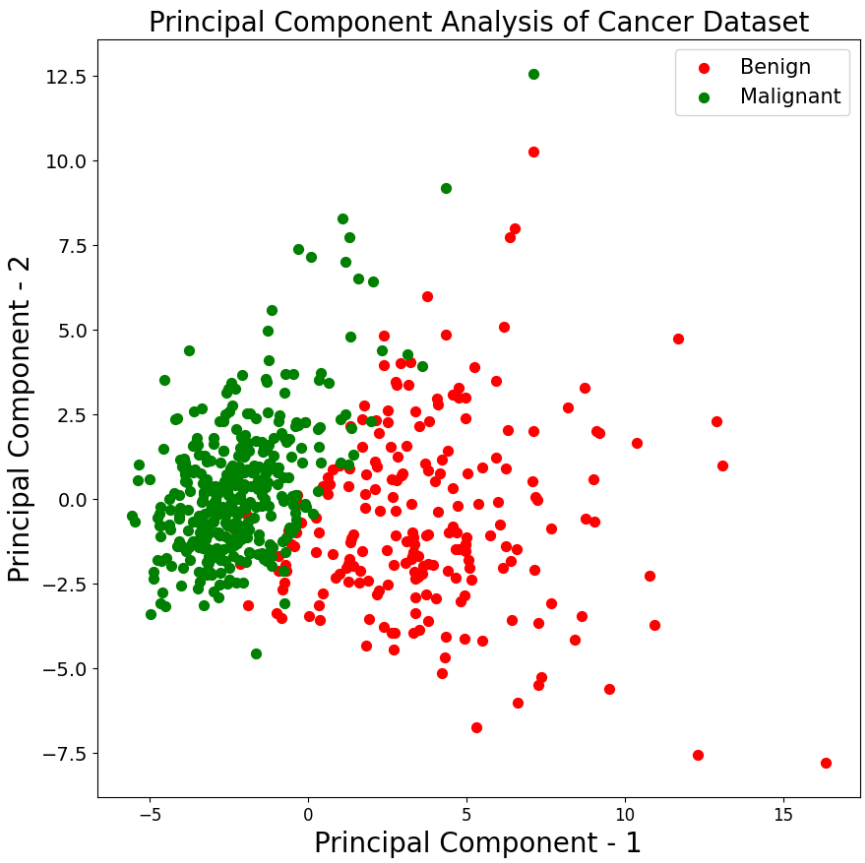
you can find the explained_variance_ratio. It will provide you with the amount of information or variance each principal component holds after projecting the data to a lower dimensional subspace.

```
print('Explained variation per principal component: {}'.format(pca_cancer.explained_variance_ratio_))
```

```
Explained variation per principal component: [0.44272026 0.18971182]
```

Plot the PCA

```
import matplotlib.pyplot as plt
plt.figure()
plt.figure(figsize=(10,10))
plt.xticks(fontsize=12)
plt.yticks(fontsize=14)
plt.xlabel('Principal Component - 1',fontsize=20)
plt.ylabel('Principal Component - 2',fontsize=20)
plt.title("Principal Component Analysis of Cancer Dataset",fontsize=20)
targets = ['Benign', 'Malignant']
colors = ['r', 'g']
for target, color in zip(targets,colors):
    indicesToKeep = cancer_dataset['label'] == target
    plt.scatter(principal_cancer_Df.loc[indicesToKeep, 'principal component 1']
               , principal_cancer_Df.loc[indicesToKeep, 'principal component 2'], c = color, s = 50)

plt.legend(targets,prop={'size': 15})
```

```
<matplotlib.legend.Legend at 0x7964e405a7a0>
<Figure size 640x480 with 0 Axes>
```